

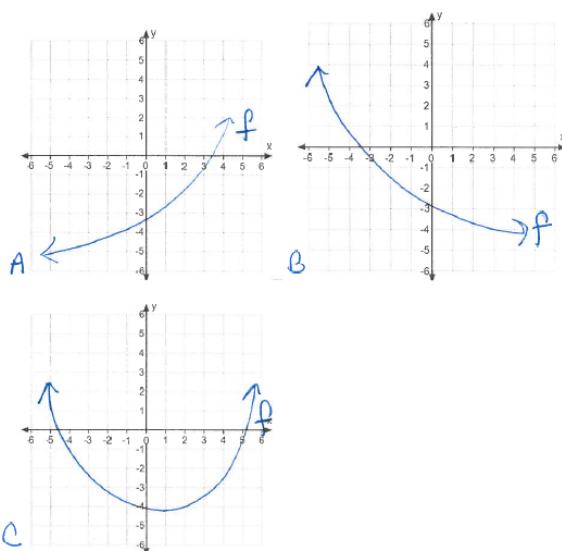
Problem 3

Problem 3

Problem 1 Give an example or sketch of a function that is continuous on $(-\infty, \infty)$ and satisfies given properties. If such a function does not exist, explain why.

- (a) A function f is concave up and negative everywhere.

Explanation. This is not possible. If a function is always concave up then at some point, the function must cross the x -axis and become positive. See the three example figures below.



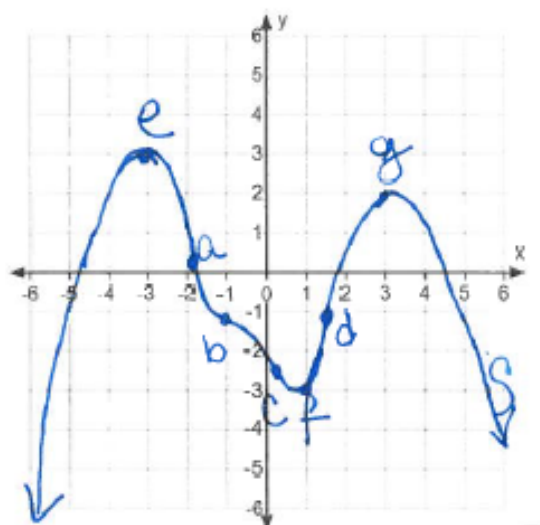
- (b) A function f is decreasing and concave up everywhere.

Explanation. This is possible. See figure B above.

- (c) A function s has exactly 3 local extrema and four inflection points.

Explanation. This is possible. See the figure below. The graph of s has inflection points at a, b, c, d and local extrema at e, f, g .

Problem 3



(d) A function f has exactly 2 zeros and one local extrema.

Explanation. This is possible. See the figures below for examples.

